Day 122: Gaussian Mixture Models and the EM Algorithm From Scratch

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 18, Day 4 | Complete study guide

Watch: Clustering (4): Gaussian Mixture Models and EM (Alexander Ihler) | Gaussian Mixture Models (GMM) & EM Algorithm Explained Visually (PhyLosophy)

Script: `day122_gmm_em.py` · every number in this guide is copied by program from `day122_run_output.txt`, the real output of that script.

A Gaussian mixture describes the data as k bell-shaped clouds, each with its own centre, shape and share, and gives every point a probability of belonging to each. Fitted by EM from scratch, it matches scikit-learn to six decimals and never lowers its likelihood in 3454 steps. Where clusters differ in shape or size it clearly beats k-means: iris 0.9039 against 0.6201, a wide-plus-tight pair 0.8988 against 0.2036. Two honest limits: BIC chose the wrong number of clusters on both real datasets, and on digits full covariances only work with an added ridge, whose best value depends on what you ask for.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

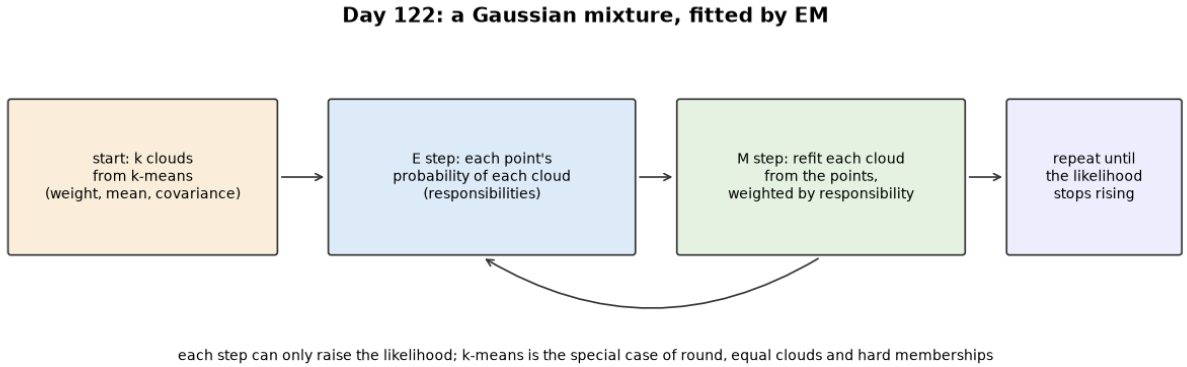
A **Gaussian mixture model** (GMM) says each point came from one of k Gaussian clouds: pick cloud j with probability w_j (its **weight**), then draw the point from a normal distribution with mean μ_j and **covariance** Σ_j (its size, stretch and tilt). Each point gets a **responsibility** for each cloud: the probability that it came from there.

EM (expectation-maximisation) fits it: the E step computes responsibilities from the current clouds; the M step refits the clouds from the responsibilities. k-means (Day 119) is the special case of round, equal clouds and all-or-nothing memberships.

REAL-WORLD SCENARIO

Heights in a mixed school: a blend of two bell curves, one for younger and one for older pupils. A height of 150 cm is not 'in' one group; it is, say, 70% likely to belong to the younger one.

SYSTEM DIAGRAM



Start from k-means; alternate the E step and the M step until the likelihood stops rising.

Today's part for the week's build

`day122_exercise.py` is the density, log-sum-exp, the E step and the M step, each checked against a formula. Its real output:

ACTUAL OUTPUT

```
log N((2, 0) | (1, -1), S): -2.634006; textbook formula with the inverse and determinant:
    -2.634006
logsumexp of [1000, 1000]: 1000.693147 (exp(1000) overflows; the answer is 1000 + log 2 =
    1000.693147)
hard memberships: weights [0.6667, 0.3333] (40 and 20 of 60); mean of group 2 equals its own mean
    True; its covariance equals np.cov(bias=True) True
responsibilities sum to 1 on every row: True; log-likelihood over 5 EM steps never falls: True
```

Data: iris and wine (standardised) and the digits, bundled with scikit-learn, labels used only to judge; and Day 121's shapes, made on purpose. scikit-learn's GaussianMixture is an outside check.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. By hand: one EM step on six points

DEFINITION — start here, no prior knowledge assumed

Points 0, 1, 2, 6, 7, 8 and a poor start: two clouds with means 1 and 5 and variance 1. One E step, one M step.

REAL-WORLD SCENARIO

Two shepherds who each claim the sheep nearest to them, then each walks to the middle of the sheep they claimed.

MATHEMATICS

$$E\ step:\ r_{ij} = w_j\, N(x_i \mid \mu_j, \Sigma_j) \,/\, \Sigma_l\, w_l\, N(x_i \mid \mu_l, \Sigma_l)$$

$$M\ step:\ N_j = \Sigma_i\, r_{ij};\quad w_j = N_j/n;\quad \mu_j = \Sigma_i\, r_{ij}\, x_i \,/\, N_j;\quad \Sigma_j = \Sigma_i\, r_{ij}\, (x_i - \mu_j)(x_i - \mu_j)^T \,/\, N_j$$

IMPLEMENTATION

```
def log_gauss(X, mu, S):
    L = np.linalg.cholesky(S)
    Z = np.linalg.solve(L, (X - mu).T)
    return -0.5 * (Z ** 2).sum(0) - np.log(np.diag(L)).sum() - 0.5 * X.shape[1] * LOG2PI
def logsumexp(A):
    m = A.max(1, keepdims=True)
    return (m + np.log(np.exp(A - m).sum(1, keepdims=True)))[:, 0]
def e_step(X, w, M, S):
    logp = np.stack([np.log(w[j]) + log_gauss(X, M[j], S[j]) for j in range(len(w))], 1)
```

```
    ll = logsumexp(logp)
    return np.exp(logp - ll[:, None]), float(ll.sum())
def m_step(X, R, kind="full", reg=1e-6):
    Nk = R.sum(0) + 1e-300; w = Nk / len(X); M = (R.T @ X) / Nk[:, None]; d = X.shape[1]; S = []
    for j in range(R.shape[1]):
        D = X - M[j]; Sj = (R[:, j, None] * D).T @ D / Nk[j]
        if kind == "diag":
            Sj = np.diag(np.diag(Sj))
        elif kind == "spherical":
            Sj = np.eye(d) * np.trace(Sj) / d
        S.append(Sj + reg * np.eye(d))
    return w, M, np.array(S)
```

ACTUAL OUTPUT

```
start: weights 0.5 and 0.5, means 1 and 5, variances 1 and 1
E step: P(left cloud | x) = [1.0, 0.9997, 0.982, 0.0, 0.0, 0.0]; log-likelihood -17.6540
by hand for x = 2: exp(-(2-1)^2/2) against exp(-(2-5)^2/2), so 1 / (1 + exp(-4)) = 0.9820
M step: weights [0.4969, 0.5031], means [0.994, 6.9695], variances [0.6647, 0.8148]
the next E step: log-likelihood -11.4863, up from -17.6540; the right cloud moved from 5 to the
    middle of 6, 7 and 8
Actual output from this lesson's run.
```

What it means

The point 2 is 0.982 likely to belong to the left cloud: by hand $1/(1 + e^{-4}) = 0.9820$. After one M step the right cloud's mean jumps from 5 to 6.9695, the middle of 6, 7 and 8, and the log-likelihood rises from -17.6540 to -11.4863. Every number is a weighted average; nothing is chosen by hand except the start.

WHY CHOOSE THIS?

Closed-form updates, simple enough to check by hand.

WHY NOT THIS? (TRADEOFFS)

Like k-means, EM finds a local optimum that depends on the start.

2. Correctness

DEFINITION — start here, no prior knowledge assumed

The log-likelihood is tracked over thousands of EM steps; then iris is fitted here and by scikit-learn from the same starting clouds.

REAL-WORLD SCENARIO

Weighing a parcel on two scales.

ACTUAL OUTPUT

```
iris and wine, 3 covariance types, 10 seeds each: 3454 EM steps; steps where the log-likelihood
    fell: 0
iris, full covariances, the same start: mean log-likelihood -1.936874 here, -1.936874 in
    scikit-learn; largest gap between the means 8.3e-07; labels agree (ARI) 1.0000
Actual output from this lesson's run.
```

What it means

The log-likelihood fell on 0 of 3454 steps, as the theory promises. From the same start, the mean log-likelihood matches scikit-learn to six decimals (-1.936874), the means agree to 8.3e-07, and the labels agree exactly (1.0000).

WHY CHOOSE THIS?

Both the guarantee and the numbers are checked.

WHY NOT THIS? (TRADEOFFS)

Matching from one start says nothing about which start is best.

3. Where a mixture beats k-means

DEFINITION — start here, no prior knowledge assumed

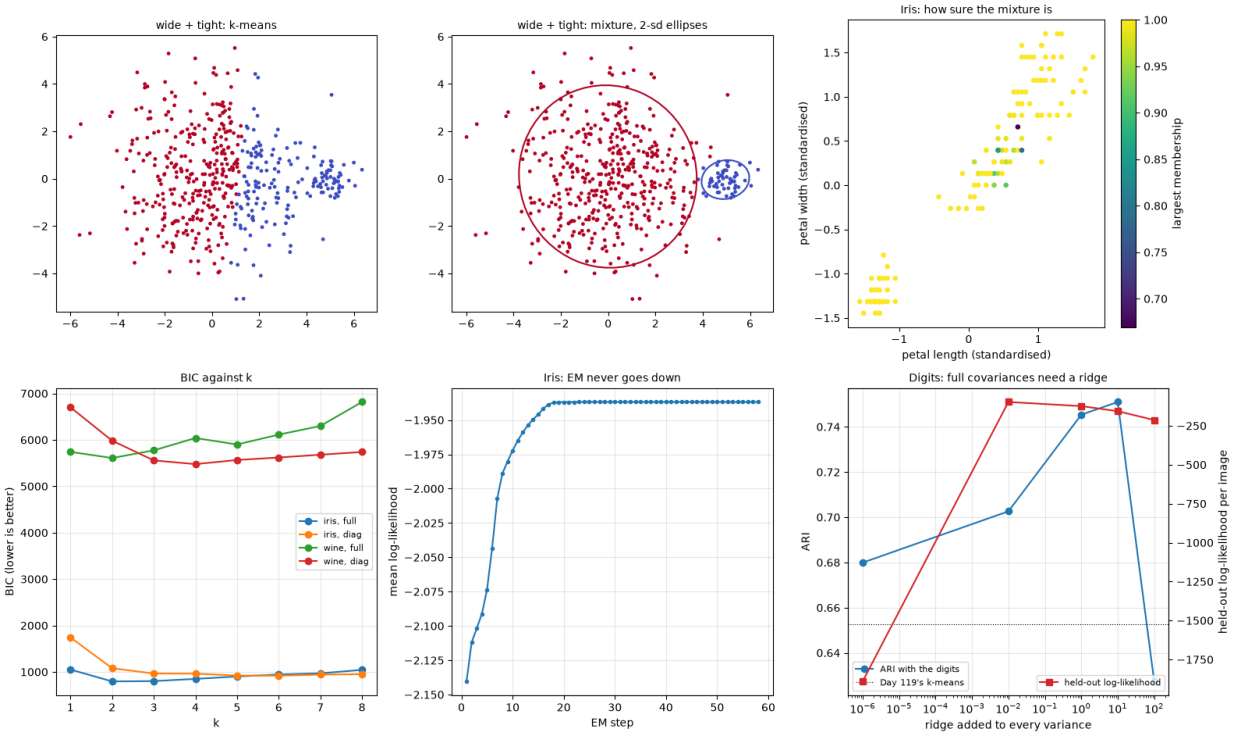
Full covariances, k set to the true number of groups, on two real datasets and two made-up shapes.

REAL-WORLD SCENARIO

Drawing an oval around each flock instead of a circle of the same size around every flock.

ACTUAL OUTPUT

```
  iris         : ARI mixture 0.9039, k-means 0.6201; cluster sizes mixture [55, 50, 45], k-means [53, 50, 47]
  wine         : ARI mixture 0.9459, k-means 0.8975; cluster sizes mixture [68, 62, 48], k-means [65, 62, 51]
  wide + tight : ARI mixture 0.8988, k-means 0.2036; cluster sizes mixture [442, 58], k-means [325, 175]
  two moons    : ARI mixture 0.2755, k-means 0.2475; cluster sizes mixture [221, 79], k-means [161, 139]
each mixture cloud has its own size and shape, so a wide group and a tight one can sit side by side; moons are
    not ellipses at all
Actual output from this lesson's run.
```



Top: k-means and the mixture on the wide-plus-tight pair (2-standard-deviation ellipses), and how sure the mixture is on iris. Bottom: BIC against k, the log-likelihood over EM steps, and the ridge on digits.

What it means

On iris the mixture reaches 0.9039 against k-means' 0.6201: two of the species are elongated, overlapping clouds that a round boundary cuts badly. On the wide-plus-tight pair, k-means splits the wide group to balance distances ([325, 175]); the mixture gives the wide group a wide cloud and finds sizes [442, 58] (true 450 and 50). The moons are not ellipses, so neither works (0.2755); that is DBSCAN's territory (Day 121).

WHY CHOOSE THIS?

Clusters of different sizes, stretches and tilts.

WHY NOT THIS? (TRADEOFFS)

Each cloud is still an ellipse; curved groups need something else.

4. Soft assignments: how sure is each point?

DEFINITION — start here, no prior knowledge assumed

The largest responsibility of each iris flower, in three bands, against whether the flower's cloud is named correctly.

REAL-WORLD SCENARIO

A doctor who says '95% sure' should be wrong about one time in twenty.

ACTUAL OUTPUT

```
  largest membership in [0.99, 1.00]: 131 flowers; named correctly 129 (0.985)
  largest membership in [0.90, 0.99):  16 flowers; named correctly 14 (0.875)
  largest membership in [0.00, 0.90):   3 flowers; named correctly 2 (0.667)
the less sure the mixture is about a flower, the more often it is wrong: its memberships say which
    assignments to trust
Actual output from this lesson's run.
```

What it means

Of the 131 flowers the mixture is at least 99% sure about, 129 are right (0.985). Below 90% it is right on 2 of 3. The less sure it is, the more often it is wrong: the probabilities carry real information that hard labels throw away.

WHY CHOOSE THIS?

Doubtful points can be flagged for a closer look.

WHY NOT THIS? (TRADEOFFS)

On 150 flowers the uncertain bands are small, so these rates are rough.

5. Choosing k with BIC

DEFINITION — start here, no prior knowledge assumed

The Bayesian information criterion trades fit against size: more clouds always fit better, but each costs parameters.

REAL-WORLD SCENARIO

Judging a story by how well it explains the facts, minus a penalty for every extra character it needs.

MATHEMATICS

$$BIC = -2 \log L + p \log n; \quad p = (k-1) + kd + k \times (d(d+1)/2 \text{ for full, } d \text{ for diagonal})$$

ACTUAL OUTPUT

```
BIC = -2 log-likelihood + (number of parameters) x log n: a better fit, minus a price for every
    extra parameter
  iris  full : best k 2 (true 3); BIC 1051, 795, 802, 849, 901, 943, 969, 1043
  iris  diag : best k 6 (true 3); BIC 1743, 1078, 965, 961, 922, 918, 941, 950
  wine  full : best k 2 (true 3); BIC 5741, 5608, 5773, 6039, 5900, 6112, 6298, 6816
  wine  diag : best k 4 (true 3); BIC 6702, 5982, 5557, 5476, 5566, 5619, 5679, 5739
iris, full covariances: 44 parameters at k = 3; wine: 314 (13 features, 178 rows)
Actual output from this lesson's run.
```

What it means

BIC did not find the true k = 3 anywhere. With full covariances it chose 2 on both datasets; wine's full model needs 314 parameters for 178 rows, a heavy penalty for every extra cloud. With diagonal covariances it chose 6 for iris and 4 for wine. BIC measures how well clouds describe the density, and the species are not clean Gaussian clouds; on iris the full-covariance BIC at k = 2 and 3 is close (795 and 802).

WHY CHOOSE THIS?

A principled number that needs no labels.

WHY NOT THIS? (TRADEOFFS)

It answers 'which density fits best', which need not be 'how many groups'.

6. Covariance shapes

DEFINITION — start here, no prior knowledge assumed

Full covariances (any ellipse), diagonal (ellipses along the axes) and spherical (circles) on iris and wine.

REAL-WORLD SCENARIO

Fitting clothes from one size per person, from separate chest and leg sizes, or from a full tailor's measurement.

ACTUAL OUTPUT

```
  data   covariance       ARI       BIC  parameters
  iris   full          0.9039       802          44
  iris   diag          0.7592       965          26
  iris   spherical     0.6199      1227          17
  wine   full          0.9459      5773         314
  wine   diag          0.8636      5557          80
  wine   spherical     0.8786      5709          44
  Day 119's k-means: iris 0.6201, wine 0.8975
spherical clouds with soft memberships behave much like k-means; full ones can tilt and stretch
Actual output from this lesson's run.
```

What it means

Full covariances win on agreement with the truth (0.9039 and 0.9459). Spherical clouds land beside k-means, as they should: 0.6199 against 0.6201 on iris. But BIC prefers the diagonal model on wine (5557 against 5773): with 13 features and 178 rows, the full model's extra parameters cost more than they gain in fit.

WHY CHOOSE THIS?

Full covariances capture tilt and correlation.

WHY NOT THIS? (TRADEOFFS)

Parameters grow with the square of the number of features.

7. When a cloud collapses: digits

DEFINITION — start here, no prior knowledge assumed

Full covariances in 64 dimensions. Some pixels never change, so no covariance can be inverted; the standard fix adds a small 'ridge' to every variance. Each ridge is judged by agreement with the digits and by the log-likelihood of 30% of images held out.

REAL-WORLD SCENARIO

A ruler marked only in whole metres measuring a pencil: you must add some tolerance before the measurement means anything.

ACTUAL OUTPUT

```
3 pixels are 0 in every image, so their variance is 0 and no covariance can be inverted
  ridge 0: the Cholesky factor fails ('Matrix is not positive definite')
    ridge     ARI   held-out log-likelihood per image
    1e-06  0.6799                          -1890.53
     0.01  0.7026                            -97.56
        1  0.7453                           -124.04
       10  0.7510                           -156.71
      100  0.6274                           -213.18
Day 119's k-means scored 0.6525; the ridge adds the same small variance to every pixel, a little
    like blurring each cloud
Actual output from this lesson's run.
```

What it means

3 pixels are always 0, so with no ridge the Cholesky factor fails. A tiny ridge (1e-06) works but overfits badly: held-out log-likelihood -1890.53 per image. The held-out likelihood is best at 0.01 (-97.56); agreement with the digits is best at 10 (0.7510, above Day 119's k-means at 0.6525). The two criteria disagree: the best density model is not the best clustering.

WHY CHOOSE THIS?

The ridge makes full covariances usable in high dimensions.

WHY NOT THIS? (TRADEOFFS)

It is one more setting to choose, and its best value depends on the goal.

Interview preparation — technical questions

Q: GMM or k-means?

A: GMM when clusters differ in size or shape, or when probabilities are needed: iris 0.9039 against 0.6201.

Q: How is a GMM fitted?

A: EM: responsibilities from the clouds, clouds from the responsibilities, until the likelihood stops rising.

Q: How do you choose k?

A: BIC is the standard; here it chose 2 for iris and wine (true 3), so check against the purpose.

Q: Why does a full-covariance GMM fail on some data?

A: Constant or duplicate features make a covariance singular; add a ridge (digits: 3 constant pixels).

Q: How do you avoid underflow?

A: Work in logs: log-densities through a Cholesky factor, and log-sum-exp for the total.

Interview preparation — theoretical questions

Q: Why does EM never lower the likelihood?

A: Each step maximises a lower bound that touches the likelihood at the current parameters.

Q: How is k-means a special case?

A: Round equal clouds with variance shrinking to 0: responsibilities become 0 or 1.

Q: What do the covariance types trade off?

A: Flexibility against parameters: full 314 for wine against 80 for diagonal.

Q: Do responsibilities mean anything?

A: Here yes: at least 99% sure was right 0.985 of the time; below 90%, 2 of 3.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> A Gaussian mixture is k-means with honest uncertainty and flexible shapes. Fitted by EM, it found iris's stretched species and a wide cloud beside a tight one where k-means could not, and its probabilities said which assignments to trust. Its weak points are the same as every density model's: choosing k, and keeping covariances well behaved when there are many features.

- Matches scikit-learn: mean log-likelihood -1.936874, labels identical; 0 falls in 3454 EM steps.
- Iris 0.9039 (k-means 0.6201); wide + tight 0.8988 (0.2036).
- BIC chose k = 2 for both iris and wine (true 3).
- Digits: ridge 0.01 best on held-out likelihood, ridge 10 best on agreement (0.7510).

Suggested watch order, if starting from zero

- 1. Clustering (4): Gaussian Mixture Models and EM (Alexander Ihler) — a lecture on mixtures and EM.
- 2. Gaussian Mixture Models (GMM) & EM Algorithm Explained Visually (PhyLosophy) — a visual walk through EM on stretched clusters.

Day 123 preview

- **Principal component analysis from scratch**: the directions of greatest variance, with a real variance-explained demonstration.